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Automated disassembly-oriented knowledge graph construction for retired battery packs using a candidate entity-based relational triple joint extraction method☆

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ABSTRACT

Currently, the disassembly of retired electric vehicle battery packs relies on manpower and results in high cost, low efficiency, and poor stability. With the development of artificial intelligence, automated disassembly is an efficient method to largely reduce even completely replace human disassembly. However, the various kinds of battery packs and the uncertainty on their retired numbers and types lead to frequent changes of their disassembly processes. It is necessary to provide a method that can integrate valuable disassembly knowledge to enable automated disassembly. Thus, this study proposes an automated disassembly-oriented knowledge graph for retired battery packs which considers the properties of subassemblies (entities) and explicit physical connections/implicit associations among subassemblies (relations). A large amount of unstructured data exists regarding battery packs, such as product manuals and maintenance records, whereas the knowledge that can be available to guide the disassembly process is dispersed and sparse. To solve this, a candidate entity-based relational triple joint extraction method is developed to efficiently extract the disassembly knowledge, which consists of semantic feature learning, candidate entity recognition, and explicit/implicit relational triple identification. Finally, more than 10,000 sentences collected from multi-source unstructured texts are adopted to verify the proposed method. The experimental results demonstrate that our proposed method achieves an F1-score of 93.99% in candidate entity recognition and an F1-score of 95.6% in triple extraction. Also, the information of disassembly operations, disassembly tools, and subassembly properties can be recommended by the automated disassembly-oriented knowledge graph for retired battery packs.

1. Introduction

According to the International Energy Agency, electric vehicle (EV) sales approached 14 million in 2023, of which nearly 60% came from China. With the rapid rise of the EV market, the demand for EV batteries has continued to grow and is expected to increase four and a half times by 2030, ushering in a rapid growth stage for retired EV batteries. At the end of 2023, the Ministry of Industry and Information Technology of the People's Republic of China issued a document regarding the comprehensive management of retired EV batteries, which stipulates that manufacturers should be correspondingly responsible for battery recycling. The standardisation and large-scale recycling of EV battery packs is an unavoidable trend, and disassembly operations are inevitable [1]. The diversity of battery types caused by non-standardised designs and the complexity of disassembly processes due to interconnections among components [2], make retired battery pack disassembly a labour-

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intensive operation that is highly dependent on manual [3,4]. This results in disassembly operations that are costly, inefficient, and unstable. With the development of artificial intelligence and the rise of Industry 5.0, automated disassembly is an efficient method to largely reduce even completely replace human disassembly [5].

However, the various kinds of battery packs and the uncertainty on their retired numbers and types lead to frequent changes of their disassembly processes. To ensure reliable automated disassembly of battery packs, the synergy of battery pack characteristic knowledge (e.g., physical connections, subassembly properties) and battery pack disassembly process knowledge (e.g., disassembly operations, disassembly tools) is essential. It is necessary to provide a method that can integrate the above disassembly knowledge to enable automated disassembly [3,6,7]. Knowledge graphs, a structured knowledge management tool that graphically displays complex relationships, can provide more associative information than traditional databases [8]. In the industrial field, knowledge graphs enable knowledge integration of multiple data sources by establishing a unified and interconnected framework [9], thereby advancing knowledge management. Its powerful knowledge integration and retrieval capabilities simplify knowledge queries, and enhance task accuracy by recommending associated knowledge [10]. Therefore, this study aims to construct an automated disassembly-oriented knowledge graph (ADOKG) for retired EV battery packs, which systematically integrates the above disassembly knowledge by modelling subassemblies as nodes with properties and explicit physical connections/implicit associations among subassemblies as edges with attributes. It can provide a comprehensive knowledge search and knowledge recommendations (disassembly operations, disassembly tools, and subassembly properties) for battery pack automated disassembly decisions.

It is quite challenging to exactly acquire the above disassembly knowledge from the multi-source unstructured battery pack texts and construct ADOKG due to the uneven distribution of entities, a high volume of overlapping triples, and numerous and complex compound noun phrase entities. Specifically, (1) uneven distribution of entities indicates that a few subassemblies (e.g., screw, harness) occur more frequently in the texts of battery packs, and some subassemblies (e.g., battery pack, module) appear less often, which increases the recognition difficulty of the low-resource named entity; (2) a high volume of overlapping triples means that several subassemblies (e.g., fastener, module) act as shared nodes, forming relations with multiple other subassemblies in the texts of battery packs, which increases the difficulty of the relational triple extraction; (3) numerous and complex compound noun phrase entities reflect that most subassemblies (e.g., protective cover of battery module 3) are constructed as multi-noun units, where some nouns serve as semantic modifiers to the core noun. This leads to erroneous segmentation of entities due to entity boundary ambiguity.

Therefore, a candidate entity-based relational triple joint extraction (CE-RTJE) method is developed in this study, aiming to efficiently extract the disassembly knowledge from the texts of battery packs. Specifically, the proposed method consists of three functional modules, i.e., the encoding module, the candidate entity recognition module, and the relational triple extraction module. The first module consists of RoBERTa-wwm-ext-large (RoBERTa) and Bidirectional Long Short-Term Memory (BiLSTM) to learn semantic features, in which RoBERTa learns dynamic word-level representations and BiLSTM learns contextual information. The second module extracts candidate entities according to the Mixture of Entity Experts (MoEE) and Conditional Random Fields (CRF), in which MoEE addresses the recognition difficulty of the low-resource named entity and CRF avoids the occurrence of illogical label sequences. The third module identifies explicit physical connection/implicit association according to the novel cascade binary tagging framework (CasRel) and the Jaccard similarity, in which CasRel facilitates the relational triple extraction caused by a high volume of overlapping triples and Jaccard similarity deals with the numerous and complex compound noun phrase entities. Compared with the existing work, our contributions are listed as follows:

(1) ADOKG is established to provide knowledge recommendations (e.g., disassembly operations, disassembly tools, and subassembly properties) for the automated disassembly of retired battery packs, in which the explicit physical connections/implicit associations, subassembly properties, and connection attributes are considered.
(2) CE-RTJE is specifically developed to efficiently extract automated disassembly knowledge (i.e., battery pack characteristic knowledge and battery pack disassembly process knowledge) from texts of battery packs. It consists of three modules: an encoding module, a candidate entity recognition module, and a relational triple extraction module.
(3) Over 10,000 sentences regarding the disassembly of retired battery packs are collected from various manuals and applied to validate the proposed CE-RTJE method. The experimental results achieve an F1-score of 93.99 % in candidate entity recognition and an F1-score of 95.6 % in triple extraction, which outperforms the existing models.

The rest of this paper is organised as follows. Section 2 presents the related works. Section 3 describes the preprocessing and text analysis of the collated battery pack dataset. Section 4 details the ADOKG construction process, including the design of the ontology model and the development of CE-RTJE. Section 5 verifies the proposed CE-RTJE and explores knowledge recommendations for the automated disassembly of retired EV battery packs. Finally, Section 6 summarises the conclusions and future insights.

2. Related works

2.1. Disassembly knowledge modelling

Knowledge modelling is the process of systematising and structuring complex and discrete information. Ontology and semantic web technologies can be employed to handle large-scale data with intricate knowledge modelling architectures [11]. Ontology achieves effective knowledge management by defining the basic concepts, relations, and axioms within a domain [12]. This enables a unified formal definition of information from different sources, which is conducive to knowledge integration, reuse, and sharing [13]. In recent years, the ontology has been applied across a wide range of fields. Based on these unified definitions, the knowledge graph abstracts the essence, interrelationships, and attributes of knowledge to construct the schema layer, then specific knowledge is treated as individuals to create the data layer, thereby forming a knowledge-rich semantic graph [14,15]. It has been employed in many fields, including fault management [16,17], product design [18,19], quality management [20–24], and manufacturing and assembling [25–27].

In the field of disassembly, scholars have begun to explore the use of knowledge graphs to provide knowledge for disassembly. Zhu and Roy [28] develop a framework for disassembly information which comprises an ontology information model and a computational model, to facilitate reasoning about a realistic optimal disassembly process sequence. Chen et al. [29] propose an ontology-based approach to predefined classes, object properties, and data properties, making the information formalised and semantic to automate disassembling decisions based on cases and rules. Guo et al. [30] propose a concept of genes for retired machinery products and use the knowledge graph as a unified and informative model representation. Yu et al. [31] propose a disassembly task planning method for automobile power batteries based on ontology and partial destructive rules, which can effectively generate disassembly task schemes. Zhang et al. [32] construct a battery fault knowledge graph using information extracted from fault logs through BiLSTM to support online knowledge queries and fault inference. Wu et al. [33] use

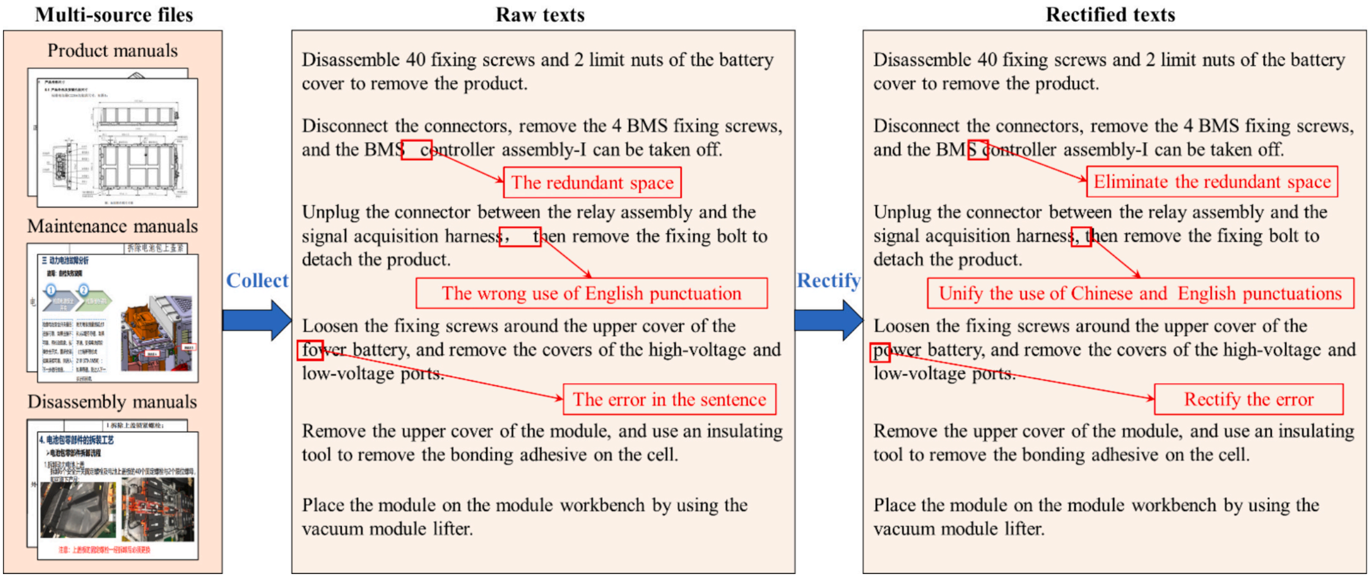


Fig. 1. An example of collecting and rectifying the dataset.

a manually constructed battery knowledge graph, which intuitively displays connections and constraint relations among components, and then they utilise topological sorting and backtracking algorithms to derive the optimal disassembly sequence.

Few scholars have conducted research from the perspective of knowledge modelling in the field of battery pack disassembling. In addition, the complex and diverse physical connections between subassemblies in retired battery packs are simplified to constraint relations in previous studies, which leads to representational limitations (e.g., specific properties cannot be assigned to entity nodes).

2.2. Knowledge graph construction

Common knowledge graph construction modes are top-down and bottom-up. The top-down construction mode carries out conceptual modelling at the schema layer first and then adds the actual data to the data layer following the predefined structure [34]. For the bottom-up mode, the schema layer of a knowledge graph is generalised from the data [35]. This study will use a combination of top-down modelling and bottom-up knowledge extraction methods to explore knowledge graph construction for retired battery pack disassembly. The top-down integration mode provides a basis for integrating subassemblies and relations in battery packs, while the bottom-up knowledge extraction method allows for the automated extraction of knowledge from multiple sources [30] and deals with the problem that complex physical associations between subassemblies are difficult to predefine at the schema layer. Now, the bottom-up knowledge extraction method mainly relies on natural language processing (NLP) technology.

Liu et al. [36] use BERT-BiLSTM-CRF model to extract wind turbine components and construct the knowledge graph by incorporating the inherent dependencies from BOM. Li et al. [37] use ALBERT-BiLSTM-CRF model to recognise fault diagnosis entities and establish relations through syntax analysis. Hu et al. [38] use BERT-BILSTM-CRF model for entity recognition of the process text and use dependency parsing to obtain the relations and attribute information among different entities to build a multi-modal process knowledge graph for wind turbines. Shen et al. [11] use NER-SOD model to extract entities from a specific customised order process information and quality inspection information. Guo et al. [30] adopt BiLSTM-CRF model to recognise the named entity of failure detection documents and product description documents and form the gene knowledge graph for a retired mechanical product.

Knowledge graph construction encompasses named entity recognition and relation extraction. However, most researches in the field of disassembly focus on automatic named entity recognition, neglecting the joint extraction of entities and relations [32]. This approach cannot deal with the problem of dispersed and sparse information in unstructured text about battery packs.

2.3. Named entity recognition

Early named entity recognition tasks were based on rules or dictionaries. With the improvement in the ability of computers to process natural language, information extraction has become more efficient [39]. Scholars have begun to explore the application of natural language processing in automatic information extraction to help knowledge graph construction [15]. Named entity recognition aims at identifying entities such as screws and cells in the text, which is usually treated as a sequence labelling problem [40]. Huang et al. [41] first applied a model that integrates BiLSTM and CRF, which uses BiLSTM to learn past and future input features and CRF to learn sentence-level labelling information. Liu et al. [42] propose a framework that consists of multi-task learning and MoEE to solve the problem of named entity recognition in low-resource target domains. The convolutional neural network, recurrent neural network (RNN), graph convolutional network, and attention mechanism have been gradually applied to named entity recognition tasks [43–46]. The BERT pre-training model has an important impact on the field of natural language processing, which enables the model to understand text semantic information well and greatly improves the performance of NER [47]. The Facebook AI team proposed Roberta based on BERT, and made the model better understand the semantic information through more training text and dynamic mask strategies [48].

2.4. Relational triple extraction

Relational triple extraction further mines the relations between entities. The current relation extraction methods based on deep learning are mainly divided into “pipeline approach” and “joint learning approach”. Early work generally adopts a pipeline approach [49], where relation extraction is regarded as a task following named entity recognition, as $f(e_1, e_2) \rightarrow R$. This strategy leads to an error propagation problem, and a joint learning strategy [50,51] is subsequently proposed, which can be specifically divided into relation extraction based on joint decoding and parameter sharing. Zheng et al. [52] convert the relation

Table 1
Detailed description, example and label for each entity.

Entity	Detailed description and example	Label
Battery pack	Provide descriptions of the battery pack, such as a 12 V lead-acid battery	B-Battery pack, I-Battery pack
Battery enclosure	Provide descriptions of the battery enclosure	B-Battery enclosure, I-Battery enclosure
Fuse box	Describe specific components in a fuse box, such as fuse and circuit breaker	B-Fuse box, I-Fuse box
Battery management system	Such as BMS master and slave board, voltage and current measurement unit	B-BMS、I-BMS
Cooling system	Such as cooling plate, cooling pipeline and water pipe	B-Cooling system, I-Cooling system
Module	Describe specific module information, such as the battery module 4	B-Module, I-Module
Cell	Describe specific information about the battery cell, such as a single lead-acid battery	B-Cell, I-Cell
Harness	Description of wiring harnesses, such as high and low-voltage wiring harnesses	B-Harness, I-Harness
Screw	Description of screws, such as fixing screw and hexagon head screw	B-Screw, I-Screw
High voltage control box	Describe components in the high-voltage control box, such as the main positive relay and pre-charge relay	B-High voltage control box, I-High voltage control box
Others	/	O

extraction task to the traditional sequence annotation task by adopting a new annotation pattern, but they cannot handle the overlapping triples problem. Zeng et al. [53] first consider the overlapping triple problem and propose a sequence-to-sequence model with a copy mechanism to extract triples. Fu et al. [54] propose a graph convolutional networks based model which modelling text as relational graphs. Wei et al. [55] propose a novel triple learning framework, termed CasRel, which initially identifies all potential subjects and employs a relation-specific tagger to detect likely relations and corresponding objects for each subject.

The massive, multi-source, unstructured battery pack texts and the fragmented internal knowledge indicate that constructing a knowledge graph through collaborative ontology modelling and NLP technology will bring high value to improving the efficiency and automation of battery disassembling tasks.

3. Dataset description and pre-processing

3.1. Data pre-processing

The dataset utilised in this study, sourced from product manuals, maintenance manuals, and disassembly manuals of battery packs provided by different manufacturers, offers significant advantages in terms of authenticity and reliability. Thus, it holds a certain degree of representative for the development of CE-RTJE and the construction of ADOKG.

Since the dataset has a large number of sentences and irregular records, pre-processing is required before knowledge extraction, as follows:

Step 1: Collect the raw texts from multi-source files. Then manually use the editing tool in Notepad to preliminarily correct the collected text, such as replacing misidentified words, adjusting English punctuation, and removing extra spaces, which can ensure the neatness and professionalism of the dataset, as shown in Fig. 1.

Step 2: Given consideration that some disassembly operations in the texts of battery packs are presented as separate records (e.g., rotate the fastening screw as illustrated, remove battery module 3). Entities (e.g., fastening screw, battery module 3) serve solely as action targets in sentences and exhibit no explicit interactions with other entities. We first perform the named entity recognition task. Filter and organise sentences containing candidate entities for the entity recognition task. A total of 10,043 sentences are collected, including 7,878 training sentences and 2,165 testing sentences, with each sentence limited to 50 words.

Table 2
Definition of physical connection relations of retired battery packs.

Connection type	Connection attribute	Detailed description	Example
Direct connecting (*dc*)	Rigidity	Describe the relation between components and fasteners, which do not rely on any external connecting elements to fix their position	(Lower battery module, Direct connecting, Five fixing screws)
Wiring harness connecting (*hc*)	Flexibility	Describe the wiring harness connection between two components	(Pre-charge relay, Wiring harness connecting, Main positive relay)
Screw fastening (*sc*)	Rigidity	The connection between two components is fixed by a screw	(Harness bracket, Screw fastening, Battery module mounting bracket for modules 6 to 10)
Adhesive bonding (*ab*)	Flexibility	Describe the connection between two components using adhesive	(Thermal pad, Adhesive bonding, Water cooling plate of modules 1)
Welding (*wd*)	Rigidity	The connection method of the two components is welding	(Module end plate, Welding, Side plate)
Snap fastening (*sf*)	Flexibility	The two components are secured with a snap	(Low voltage wiring harness, Snap fastening, Battery tray)
Cable tie fastening (*cf*)	Flexibility	The two components are secured with cable ties	(Low voltage wiring harness of battery module 9, Cable tie fastening, Low voltage wiring harness of battery module 11)

Step 3: Remove the candidate entities whose relationships cannot be determined. This is a key step against the impact of the data noise.

Step 4: Further organise the remaining sentences to extract valid triples from them. A total of 8,480 sentences are collected, providing a solid data foundation for subsequent knowledge graph construction.

3.2. Candidate entity labelling

According to the research purposes of this study, the candidate entities are divided into 10 classes: “Battery pack”, “Battery enclosure”, “Fuse box”, “Battery management system”, “Cooling system”, “Module”, “Cell”, “Harness”, “Screw”, and “High voltage control box”. The detailed description, example, and label for each entity are shown in Table 1. These 10 types of entities represent the subassemblies in retired EV battery packs, which are composed of components and parts. Note that fasteners are considered as parts in this study.

By labelling the text, it can help the model accurately identify entities and their locations. The common entity labelling schemes are BIO, BIEOS, and BMES. Due to its simplicity and low computational resource requirements, this study adopts the BIO scheme for labelling the dataset, where “B” marks the starting position of each entity, “I” marks the middle and the end positions of each entity, and the non-entity is marked with “O”. The dataset is labelled according to specific definitions and corresponding label representations.

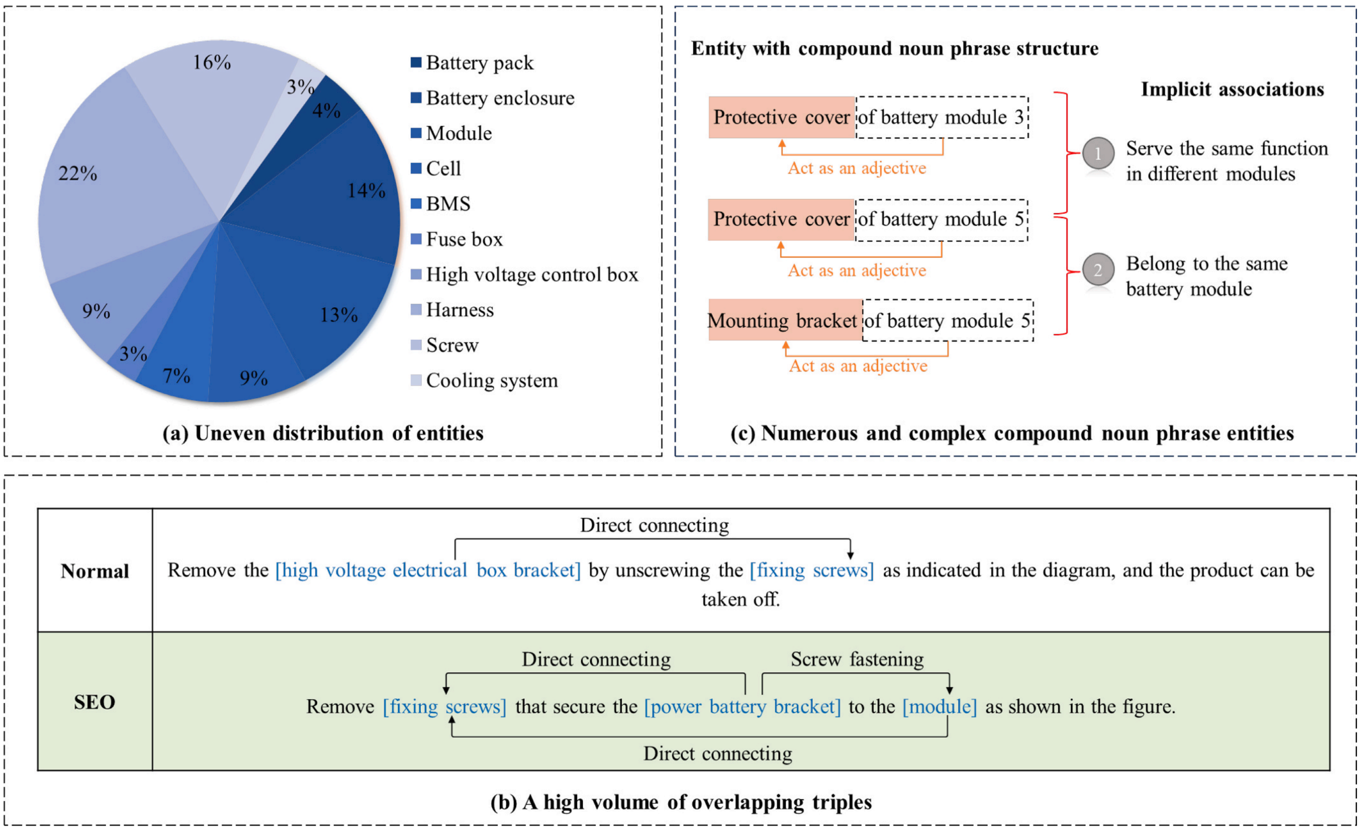


Fig. 2. Overview of text analysis.

3.3. *Triple labelling*

To address the research gap about different physical connections in retired EV battery packs, we categorise the connections between components into 6 specific types: "Wiring harness connecting", "Screw fastening", "Adhesive bonding", "Welding", "Snap fastening", and "Cable tie fastening". Furthermore, to handle the polysemy of fasteners, especially the same fasteners in different components, this study defines the connection type between fasteners and their associated components as a "Direct connecting", such as five fixing screws connecting the battery cover and tray. Additionally, two connection attributes for each connection type, rigidity and flexibility, have been defined to better support the automated disassembly.

Table 2 provides detailed descriptions, connection attributes, and examples of each connection type. During the labelling process, a total of 8,480 sentences are systematically labelled by using the Label-Studio annotation tool, strictly adhering to the predefined triple standards.

3.4. *Text analysis*

After analysing the dataset, the following characteristics are identified, as shown in Fig. 2, which pose challenges to traditional knowledge extraction methods. The corresponding approaches are detailed in Section 4.2.

(1) Uneven distribution of entities.

Due to the distributed design of fasteners, screws and wiring harnesses exhibit characteristics of high quantity and spatially dispersed distribution, leading to frequent descriptions of them during disassembly operations [3]. The "one-to-many" dependency between functional components and fasteners (e.g., the fixation of a single battery module relies on multiple screws and harnesses) further amplifies this phenomenon. Additionally, each operation is independently recorded in battery disassembly texts, ultimately resulting in an uneven distribution of entities in the battery pack texts, i.e., a few subassemblies (e.g., screw, harness) occur more frequently in the texts of battery packs, and some subassemblies (e.g., battery pack, module) appear less often. As illustrated in Fig. 2 (a), entities such as "Harness" and "Screw" occur more frequently in the dataset. The overall data exhibits a long-tail distribution, which increases the recognition difficulty of the low-resource named entity.

(2) A high volume of overlapping triples.

Within battery pack topologies, some fasteners act as shared nodes and establish relations with multiple functional components (e.g., a screw secures both a battery module and a battery bracket). As shown in Fig. 2 (b), such structural characteristics result in a large number of *SingleEntityOverlap* (SEO) triples in battery pack texts. In contrast to the triple in the normal category, which does not contain overlapping entities or relations, the triple in the SEO category involves shared entities that function as both subjects and objects across multiple triples within the same sentence. This means a high volume of overlapping triples in battery pack texts, which results in the difficulty of relational triple extraction.

(3) Numerous and complex compound noun phrase entities.

The entity with a compound noun phrase structure is very common in battery pack texts. Because through such compound noun phrase structure, entities can achieve precise engineering object definitions. For instance, in entities like "protective cover of battery module 3" and "mounting bracket of battery module 5", numerical identifiers "3"/"5" establish spatial coordinates, while "battery module" as the parent assembly defines functional ownership of "protective cover" and "mounting bracket". This means that numerous and complex compound noun phrase entities in battery pack texts, i.e., most subassemblies are constructed as multi-noun units, where some nouns serve as semantic modifiers to the core noun, as shown in Fig. 2 (c). This leads to erroneous segmentation of entities due to entity boundary ambiguity. If implicit associations between such entities (e.g., the derivation of the

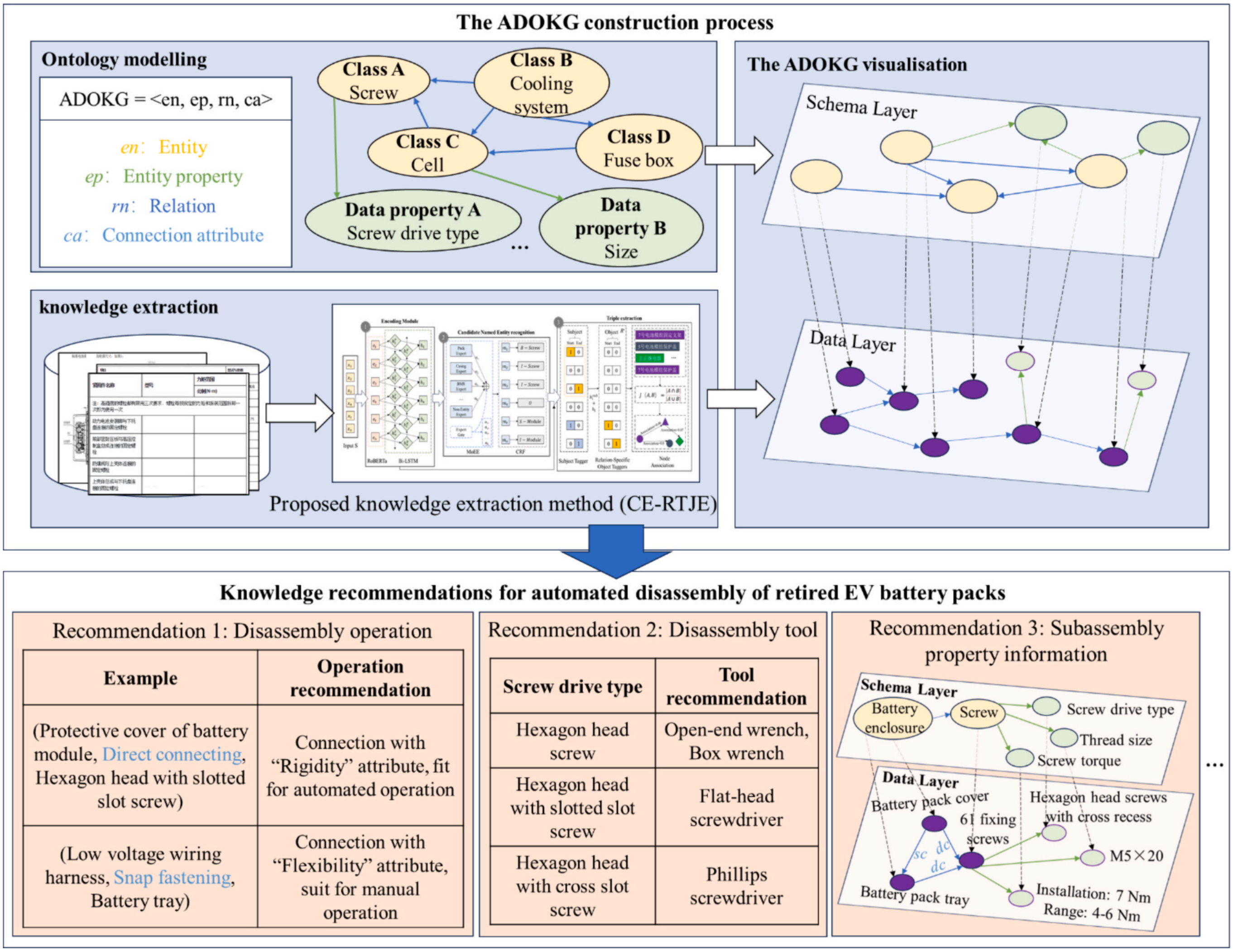


Fig. 3. The framework of ADOKG.

subassembly spatial containment relations and function similarity) are mined, not only can prevent the nodes from being discrete and ensure the accuracy and systematicity of the knowledge graph, but also contribute to the specialisation of disassembly workstations and promotes the automation of battery pack disassembly.

4. ADOKG construction

This study aims to construct ADOKG for retired EV battery packs based on the disassembly knowledge from relevant documents. The framework, as illustrated in Fig. 3, adopts the top-down ontology modelling and bottom-up knowledge extraction method to construct ADOKG. It consists of two parts: ontology modelling and knowledge extraction. Firstly, the ontology model aims to construct the schema layer of the knowledge graph, which will be described in Section 4.1. Secondly, by using the proposed knowledge extraction method described in Section 4.2, knowledge is extracted from the above dataset to construct the data layer of the knowledge graph. Ultimately, a high-precision, fine-grained ADOKG is formed, providing knowledge recommendations for the automated disassembly of retired EV battery packs.

4.1. Ontology modelling of ADOKG

Ontology is a unified and standardised method for describing domain knowledge, supporting the formal semantic representation of concepts. This study designs an ontology model of ADOKG from the perspective of automatic disassembly, providing a structured definition for its construction, as shown in Fig. 4. The ontology model of ADOKG is developed in a top-down manner, which predefines the entities corresponding to the subassemblies, and the relations corresponding to the explicit physical connections and implicit associations among subassemblies. Also, it defines entity properties and connection attributes to provide richer knowledge recommendations for automated disassembly.

Wherein, *en* represents entities in the knowledge graph, which have been categorised into 10 classes (described in Chapter 3.2) covering internal subassemblies of retired battery packs. *rn* denotes relations between these entities, which can be defined as 7 object properties based on Chapter 3.3, reflecting how subassemblies are connected. We further added an "Association" relation to describe the implicit association among subassemblies. *ep* is used to reflect the property or characteristic of an entity, which is defined as a data property. Through the use of *ep*, we can integrate product specifications provided by manufacturers to describe entities with fine-grained details, better supporting the automated disassembly of retired EV battery packs. Different entities possess distinct properties. For example, the cell entity has properties such as battery type, shape, and size, whereas the screw entity has properties like thread size, screw drive type, and screw torque. *ca* is attributes that describe the explicit physical connections. This study proposes two attributes: rigidity and flexibility, which can determine whether disassembly is performed by manual operation, mechanised operation, or

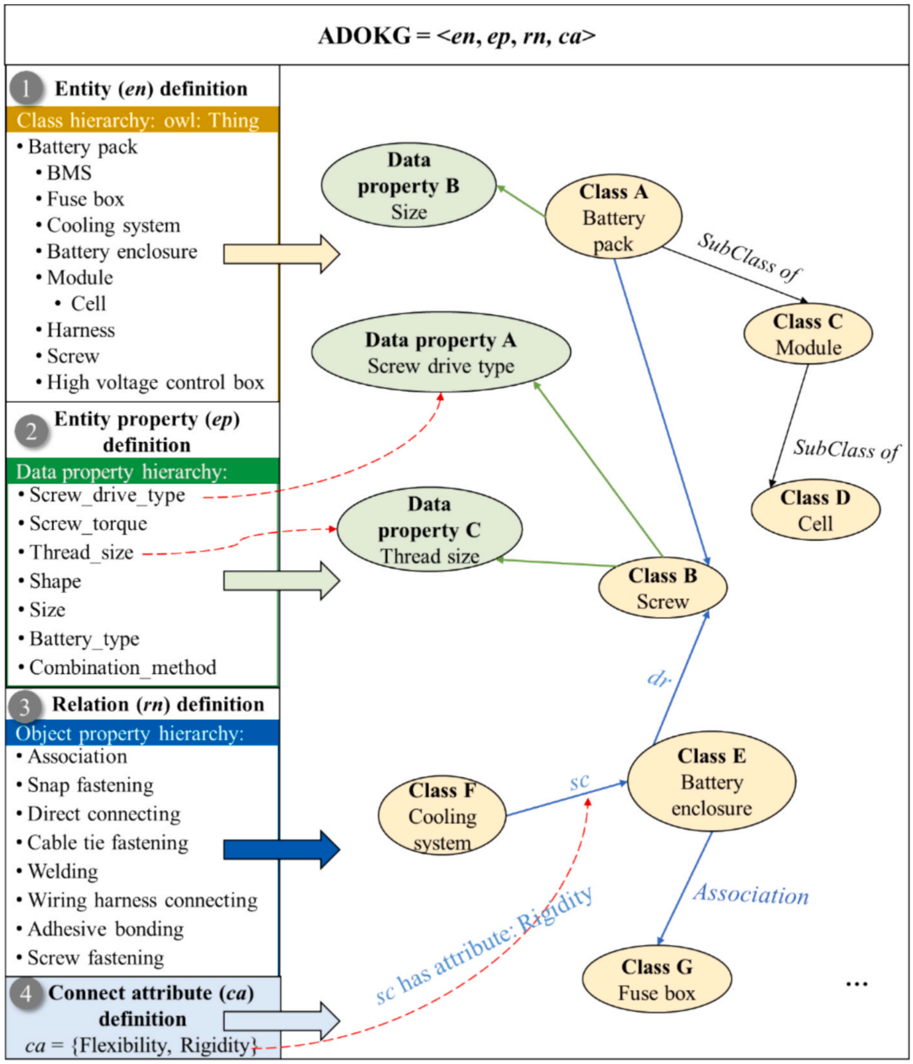


Fig. 4. The ontology model of ADOKG.

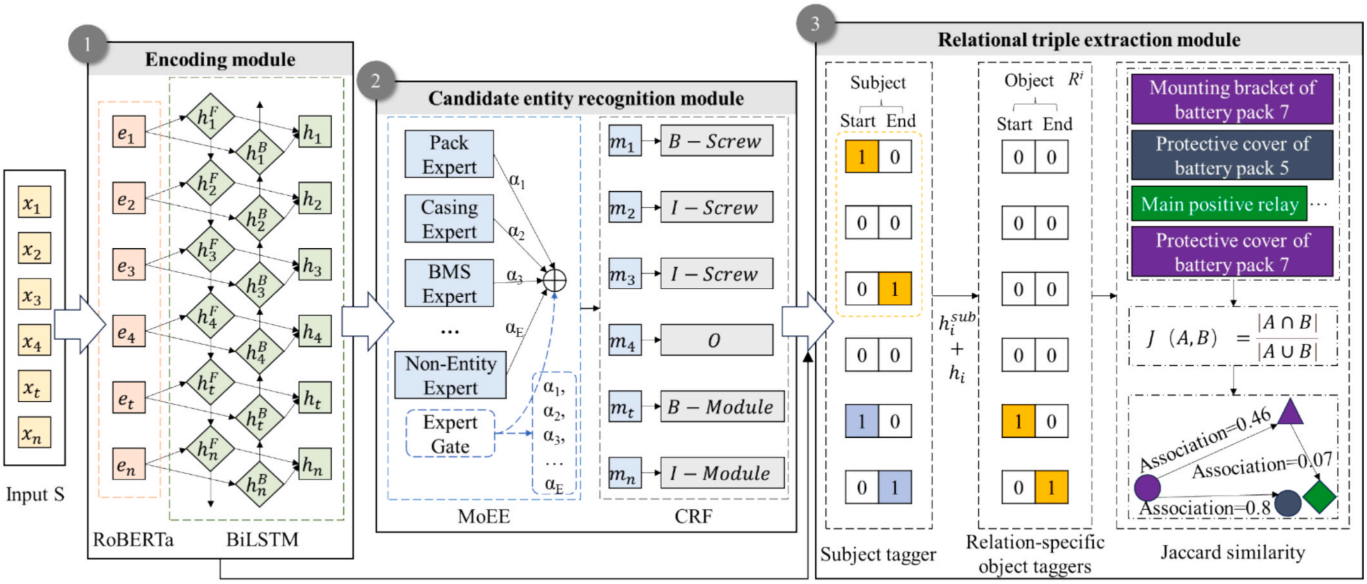


Fig. 5. The overall architecture of CE-RTJE.

automated operation.

However, the top-down approach is not necessarily complete, as it is not flexible to address various specific relations among subassemblies. Therefore, this study will incorporate a bottom-up approach by extracting specific physical connections and mining implicit associations between subassemblies from the dataset to construct the knowledge graph.

4.2. CE-RTJE

Our study has designed CE-RTJE, which aims to extract explicit physical connections and implicit associations among subassemblies in the form of relational triples to automatically construct the data layer of ADOKG, where subjects and objects are *en* and relations are *rn*. The overall architecture of CE-RTJE can be divided into three modules: the encoding module, the candidate entity recognition module, and the relational triple extraction module, as illustrated in Fig. 5. (Specifically, see Section 4.2.1, Section 4.2.2, and Section 4.2.3) .

In the encoding module, we integrate RoBERTa and BiLSTM, which enables CE-RTJE to enhance the understanding of deep semantic features in the texts. The text is input into RoBERTa to obtain dynamic word-level embeddings and fed into BiLSTM for bidirectional encoding.

Based on the word embeddings obtained from the encoding module, we combine MoEE and CRF to perform the candidate entity recognition task. The incorporation of MoEE allows CE-RTJE to learn the characteristics of different entity classes, effectively addressing the challenge caused by the uneven distribution of entities. CRF ensures a reasonable label sequence by learning the dependencies between labels, which is commonly used in the named entity recognition task.

The relational triple extraction module identifies all possible (subject, relation, object) triples in a sentence. We integrate the idea from CasRel, which can effectively deal with the challenge caused by a high volume of overlapping triples. It transforms the relation extraction task into extracting five components (s_h, s_t, *rel*, o_h, o_t), wherein s_h and s_t represent the start and end positions of the subject, while o_h and o_t indicate the start and end positions of the object, and *rel* means relations. Traditional CasRel relies on Bidirectional Encoder Representations from Transformer (BERT) for feature learning, but we improve it by substituting BERT with our custom encoding module. Moreover, to identify implicit associations among subassembly entities, we introduce the Jaccard similarity to capture associations among entities that have a compound noun phrase structure.

4.2.1. Encoding module

(1) RoBERTa for learning semantic feature.

RoBERTa is a pre-training model developed based on BERT. Compared to one-hot encoding, word2vec, or other pre-training models, RoBERTa inherits the advantage of BERT, enabling the model to understand semantic features more effectively. But unlike BERT, which uses single characters as units for masking, RoBERTa masks all characters within a word. This approach better preserves the integrity of semantics and is more suitable for our knowledge extraction task. It allows CE-RTJE to obtain dynamic word-level vector representations. RoBERTa used in our research consists of 24 layers of transformer encoders, 16 self-attention heads, and 1024 hidden units. The input sentence S ($S = \{x_1, x_2, x_3, \ldots, x_n\}$, x_i represents the *i-th* character) is converted to the initial vector representation $\{e_1, e_2, e_3, \ldots, e_n\}$.

(2) BiLSTM for learning contextual feature.

The Long Short-Term Memory (LSTM) network is an improved version of RNN. It can overcome the challenge that RNN faces when processing lengthy textual data. BiLSTM consists of two separate LSTM networks, as shown in Fig. S1 of the **Supplementary Files**. Compared with a single LSTM, the main advantage of BiLSTM is its ability to capture complete context information at each timestep within a sequence. Consequently, this enables CE-RTJE to acquire richer contextual features, thereby enhancing its semantic comprehension capabilities.

4.2.2. Candidate entity recognition module

(1) MoEE for addressing the uneven distribution of entities.

Due to the uneven distribution of entities in texts of battery packs, the traditional model, such as BiLSTM, faces the recognition difficulty of the low-resource named entity. By introducing MoEE, which is specifically designed for recognition difficulty in low-resource named entities, CE-RTJE acquires the ability to handle the uneven distribution of entities. We set an entity expert for each entity class in Section 3.2 (non-entity class “O” being treated as a special entity category), which consists of a linear network layer. An expert gate is also been set up, which consists of a linear layer followed by a softmax layer, generating the confidence distribution of each entity network. By combining different output weights, the final output of this module can synthesise the opinions of different experts. A visual representation of MoEE can be found in Fig. S2 of the **Supplementary Files**, and the definition of MoEE is as follows:

$$[exp_1(h_i), exp_2(h_i), exp_3(h_i), \ldots, exp_E(h_i)] = [L_1(h_i), L_2(h_i), L_3(h_i), \ldots, L_E(h_i)] \tag{1}$$

$$[\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_E] = Softmax(Linear(h_i)) \tag{2}$$

$$m_i = \sum_{a=1}^{E} \alpha_a * exp_a(h_i) \tag{3}$$

where L denotes the linear layer, h_i represents the input features from the encoding module, $exp_t(h_i)$ refers to the features generated by the *t-th* expert and α_i is the weight of the *i-th* expert.

(2) CRF for determining optimal prediction sequence.

The word label vectors produced by MoEE are then input into CRF, which can determine the optimal prediction sequence by considering the relationship between neighbouring labels. This network is commonly used in named entity recognition tasks to circumvent the occurrence of illogical label sequences, ensuring that labels such as “I-Screw” do not erroneously emerge at the beginning of a screw entity.

4.2.3. Relational triple extraction module

(1) Subject tagger.

This module takes the output of the encoding module as input and identifies the position of the subject entity in the encoded sentence. It employs two binary classifiers to calculate the probability of each token being the start or end position of the subject entity, as depicted in Fig. S3 of **Supplementary Files**. In this study, we set the special threshold as 0.5. If the probability exceeds the specific threshold, the token is assigned a label of 1, indicating that it is considered the start or end position of the subject. Otherwise, it is assigned a label of 0. The operations of the two classifiers are as follows:

$$p_i^{sub-start} = Sigmoid(Linear(h_i)) \tag{4}$$

$$p_i^{sub-end} = Sigmoid(Linear(h_i)) \tag{5}$$

where h_i is the encoded representation of the *i-th* token in the input vector, $p_i^{sub\_start}$ is the probability that the *i-th* token is the start position of the subject entity, and $p_i^{sub\_end}$ is the probability that the *i-th* token is the end position of the subject entity.

(2) Relation-specific object taggers.

In this study, potential object entities under a certain relation are extracted by using two binary classifiers that calculate the probability of each token being the start or end position of the object entity. As such, we set the special threshold as 0.5. If the probability exceeds the specific threshold, the token is assigned a label of 1, indicating it is considered the start or end position of the object entity. Otherwise, it is assigned a label of 0. Unlike the subject tagger, which relies on the sentence vector encoded by the encoding module as input, the relation-specific object taggers are considered more comprehensive in features. They take into consideration both the start and end features of the subject as well as the sentence features, as depicted in Fig. S4 of **Supplementary Files**. The detailed operations of the relation-specific object tagger are as follows:

$$h_i^{input} = \frac{\left(h_i^{sub\_head\_pos} + h_i^{sub\_tail\_pos}\right)}{2} + h_i \tag{6}$$

$$p_i^{obj\_start} = Sigmoid\left(Linear\left(h_i^{input}\right)\right) \tag{7}$$

$$p_i^{obj\_end} = Sigmoid\left(Linear\left(h_i^{input}\right)\right) \tag{8}$$

where $p_i^{obj\_start}$ and $p_i^{obj\_end}$ represent the probabilities of identifying the *i-th* token in the input vector as the start and end positions of the object entity, respectively. $h_i^{sub\_head\_pos}$ and $h_i^{sub\_tail\_pos}$ respectively denote the start and end features of the subject entity. Thus, h_i^{input} is the fused feature which inputs into the relation-specific object tagger.

For the explicit relational triple extraction, this study proposes that the loss function consists of two parts, weighted subject recognition loss and relation-specific object recognition loss. The weighted subject recognition loss is the product of the cross-entropy loss at the start and end positions of the subject and a weight coefficient (set as 3 in this study). The aim is to strengthen the contribution of the subject recognition task during gradient backpropagation, enhance the subject recognition ability of CE-RTJE, and supply high-quality contextual features for the relation-specific object recognition task. The specific loss function formula is as follows:

$$Total\_loss = \alpha_{sub} \times \left(C_{sub}^{start} + C_{sub}^{end}\right) + \left(C_{obj}^{start} + C_{obj}^{end}\right) \tag{9}$$

where C_{sub}^{start} and C_{sub}^{end} represent the cross-entropy loss at the start and end positions of the subject, and α_{sub} stands for the weighted coefficient of the subject recognition task. C_{obj}^{start} and C_{obj}^{end} respectively denote the cross-entropy loss at the start and end positions of the relation-specific object.

(3) Jaccard similarity calculation

Given that entity texts in this study are generally short and each consists of several noun words, when the association between two entities is strong, the corresponding entity texts will exhibit high consistency in their adjectives. Therefore, the Jaccard similarity becomes an ideal tool for inferring implicit associations between subassembly entities, as shown in Formula 10.

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{10}$$

where A and B represent texts of different entities. $\cap$ represents the intersection and $\cup$ represents the union.

This approach not only aids in establishing associations between entities but also can correct textual ambiguities arising from irregular or erroneous records, thereby enhancing the accuracy of ADOKG. In this way, a high-quality and tightly structured ADOKG can be constructed.

5. Experimental results and discussions

5.1. Experimental preliminaries

In this section, we conduct experiments to validate the effectiveness of the proposed CE-RTJE. Given the consideration that some entities serve solely as action targets in individual sentences and exhibit no explicit interactions with other entities (e.g., rotate the fastening screw as illustrated, remove battery module 3). Their relation identification has become difficult. We first perform the named entity recognition experiment on 10,043 sentences, of which 7,878 are designated as training data and 2,165 as testing data. Then we ignore these entities with insufficient relations and obtain a total of 8,480 sentences. On this basis, triple extraction is performed, with 7,840 sentences used as training data and 1,500 sentences used as testing data. Experiments are implemented in Windows 10, using Python version 3.9.19 and Pytorch 2.0 as the development language. The hardware is Intel(R) Core (TM) i9-14900 K 3.20 GHz, 128 GB memory, and NVIDIA GeForce RTX 4070 Ti GPU.

Table 3
Model parameters configuration.

Parameter description	Parameter
Sequence length	50
Dropout rate	0.5
Optimizer	Adam
Batch size in the candidate entity recognition experiment	32
Learning rate in the candidate entity recognition experiment	1e-4
Epoch in the candidate entity recognition experiment	30
Batch size in the triple extraction experiment	6
Learning rate in the triple extraction experiment	1e-3
Epoch in the triple extraction experiment	100

Table 4
Prec, Rec, and F1 of different models.

Model	Prec	Rec	F1
BiLSTM-CRF	25.34 %	28.12 %	26.59 %
Ro-BiLSTM-CRF	90.55 %	97.37 %	93.77 %
Ro-BiLSTM-MoEE-CRF	90.49 %	98.00 %	93.99 %

The model parameters (including batch size and learning rate) are predetermined based on the parameter-tuning experiments. Specifically, the batch size is tuned from 6 to 48 and every time it is increased by applying the recurrence relation $a_n = a_{n-1} + (10 + 2n)$, and the learning rate is tuned from 10^{-5} to 10^{-3} and every time increased by 10 times. Then, the best parameter settings are adopted in the proposed method, which are presented in Table 3. The maximum length of each sentence is set to 50, the dropout rate is set to 0.5, and the optimiser used is Adam. For the candidate entity recognition experiment, the batch size per epoch is 32, the maximum number of training epochs is 30, and the learning rate is 0.0001. The batch size for the triple extraction experiment is set to 6 and the epoch value is 100.

Evaluation indicators adopted in experiments are Precision (Prec), Recall (Rec), and F1-score (F1). The F1 is the harmonic mean of Prec and Rec. These indicators are defined as follows:

$$\text{Prec} = \frac{TP}{TP + FP} \tag{11}$$

$$\text{Rec} = \frac{TP}{TP + FN} \tag{12}$$

$$\text{F1} = \frac{2 * Prec * Rec}{Prec + Rec} \tag{13}$$

where TP represents the number of samples that are truly positive and correctly identified as such, FP denotes the number of negative samples mistakenly predicted as positive samples (also known as Type I error), and FN represents the number of positive examples that are incorrectly classified as negative (also known as Type II error).

5.2. Candidate entity recognition results

Since the candidate entity recognition module in our proposed CE-RTJE is optimised by the commonly employed BiLSTM-CRF model, we have selected it as the initial model. By incrementally introducing the RoBERTa and MoEE modules to evaluate their contributions and validate the effectiveness of our method in the candidate entity recognition task. The results are shown in Table 4. Based on the initial model, the

Table 5
Prec, Rec, and F1 of different models.

Model	Prec	Rec	F1
CasRel	83.8 %	90.1 %	86.8 %
BiRTE	86.8 %	85.2 %	86.0 %
ConCasRTE	88.3 %	80.9 %	84.4 %
CE-RTJE (ours)	94.9 %	96.3 %	95.6 %

incorporation of RoBERTa into the architecture to form RoBERTa-BiLSTM-CRF (Ro-BiLSTM-CRF) model resulted in a 67.18% increase in F1, significantly improving the accuracy in the candidate entity recognition task. Then, continue to add MoEE to gain RoBERTa-BiLSTM-MoEE-CRF (Ro-BiLSTM-MoEE-CRF) model, whose F1 value increased by 0.22%. These findings indicate that RoBERTa and MoEE modules can strengthen the ability of the initial model to recognise entities. It demonstrates the effectiveness of our proposed CE-RTJE method, which integrates these two modules, for the candidate entity recognition task.

5.3. *Triple extraction results*

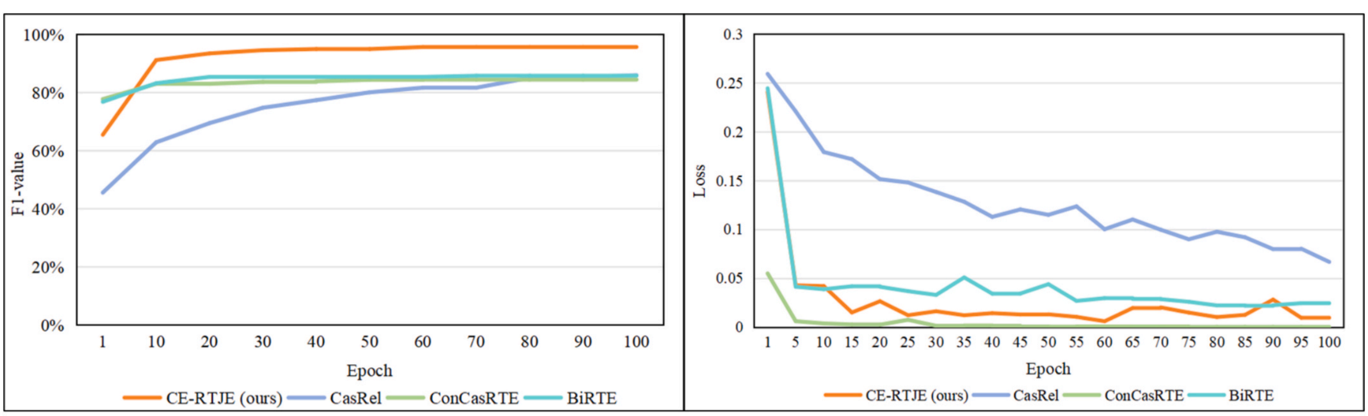
Since the relational triple extraction module in CE-RTJE is optimised on the CasRel model, we use CasRel as the compared model and conduct triple extraction experiments based on the results after eliminating relation-missing entities. BiRTE and ConCasRTE models have also been selected as comparison models to observe the performance of our method. The results are presented in Table 5. Compared with the original model CasRel, the Prec, Rec, and F1 of our proposed CE-RTJE improved by 11.1%, 6.2%, and 8.8%. The improvement effect is obvious, indicating that our proposed method can understand contextual semantic information and is effective in the relational triple extraction task. Compared with ConCasRTE and BiRTE models, the performance of CE-RTJE is still wonderful, which proves the effectiveness of our proposed method.

As seen in Fig. 6 (a), the F1 of our proposed method exceeds that of the original CasRel in every epoch. Compared to the ConCasRTE and BiRTE models, the F1 value curve of the CE-RTJE model is significantly higher than those of the other models after 10 epochs. From Fig. 6 (b), it is evident that our proposed CE-RTJE converges faster than the original CasRel, which requires approximately 90 epochs to converge, whereas our method only takes about 30 epochs. The loss value of BiRTE is between CE-RTJE and CasRel, but its final loss remains higher than that of CE-RTJE. Although the loss value of ConCasRTE is lower compared to other models, its F1 score is average, which may indicate potential overfitting.

Furthermore, to study the performance of our method under different sizes of training sets, different numbers of training data are set. Specifically, the study utilises training sets with six distinct sizes: over 1,000, 3,000, 6,000, 9,000, 12,000, and 15,000 sentences. To maintain a consistent proportional relationship, the size of the testing set is set to 377 when the training set size only exceeds 1,000. In contrast, for all other training set sizes, the testing set size is fixed at 1,500.

Generally speaking, the performance of most models in relational triple extraction tasks is gradually improved when the training data size becomes larger, as shown in Table 6. Among them, the performance of CasRel shows the most significant improvement as the training data size increases, while the other three models exhibit minimal performance improvements. When the training data size increases from 3,000 to 6,000, these models generally get the highest improvement. After that, increasing data only brings little benefit even increase computational costs. It is obvious that, across every training data size, CE-RTJE performs better than CasRel, BiRTE, and ConCasRTE in terms of Precision, Recall, and F1-score. Notably, even with limited training data, CE-RTJE can still achieve a high F1. This proves the excellent performance of CE-RTJE, as well as its stability and superiority in handling data across various scales.

As depicted in Fig. 7, regardless of the training data size, our proposed method consistently achieves a higher F1 than the original CasRel across all epochs, and it reaches optimal accuracy with a faster convergence speed, indicating that the optimisation of the encoding



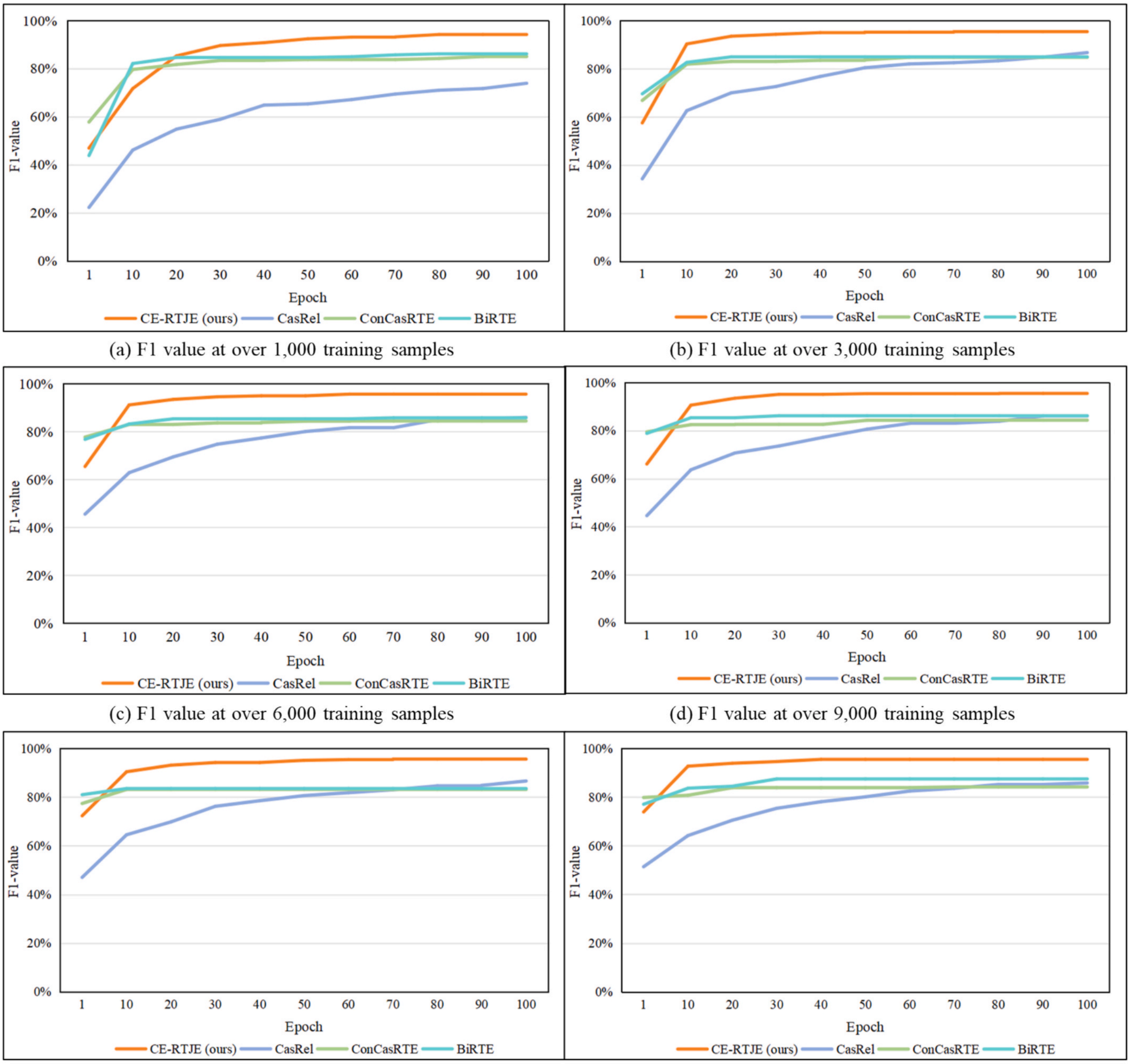
(a) F1 value of CE-RTJE (ours) and other models

(b) Loss value of CE-RTJE (ours) and other models

Fig. 6. Comparison of experimental results between our method and other models.

Table 6
Prec, Rec, and F1 of different models across various training data sizes.

Training data size	CasRel			BiRTE			ConCasRTE			CE-RTJE (ours)		
	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1
1,000	65.6 %	84.5 %	73.9 %	87.4 %	84.9 %	86.1 %	88.9 %	81.5 %	85.0 %	92.8 %	95.6 %	94.2 %
3,000	82.9 %	90.8 %	86.7 %	87.8 %	82.3 %	85.0 %	88.7 %	81.1 %	84.8 %	94.9 %	96.0 %	95.4 %
6,000	81.8 %	90.7 %	86.0 %	86.8 %	85.2 %	86.0 %	88.3 %	80.9 %	84.4 %	94.9 %	96.3 %	95.6 %
9,000	81.3 %	91.6 %	86.1 %	88.0 %	84.4 %	86.2 %	88.5 %	80.6 %	84.4 %	95.1 %	96.3 %	95.7 %
12,000	82.1 %	91.6 %	86.6 %	84.0 %	83.0 %	83.5 %	88.4 %	78.5 %	83.1 %	95.8 %	95.3 %	95.6 %
15,000	83.9 %	89.6 %	86.6 %	88.1 %	86.8 %	87.5 %	88.6 %	80.1 %	84.2 %	96.4 %	94.7 %	95.5 %



(a) F1 value at over 1,000 training samples

(b) F1 value at over 3,000 training samples

(c) F1 value at over 6,000 training samples

(d) F1 value at over 9,000 training samples

(e) F1 value at over 12,000 training samples

(f) F1 value at over 15,000 training samples

Fig. 7. Experimental results in different sizes of training samples.

Table 7
Ablation experiments.

RoBERTa	BiLSTM	Prec	Rec	F1
√	√	94.9 %	96.3 %	95.6 %
√	–	83.8 %	90.1 %	86.8 %
–	–	57.2 %	65.8 %	61.2 %

* "√" represents that the corresponding network is included. "-" represents that the corresponding network is excluded.

module in CE-RTJE has effectively improved its semantic comprehension and feature recognition abilities, resulting in a marked increase in overall performance. Compared with ConCasRTE and BiRTE, CE-RTJE achieves a higher F1 score, which proves its superior performance. Furthermore, when the training data is small (e.g., just over 1,000 sentences), our method can still maintain robust performance, achieving a significantly higher F1 than other models, as shown in Fig. 7 (a). This enhancement is attributed to our method of comprehensively utilising the information in the training data, encompassing both semantic and contextual information, thereby enhancing its generalisation capabilities.

To better understand the contribution of different networks to the last performance, we conduct ablation studies, which are illustrated in Table 7. First, CasRel can be viewed as an ablation study of CE-RTJE eliminates the BiLSTM network. Without the BiLSTM network in CE-RTJE, CE-RTJE suffers a performance degradation, about a drop of 8.8% in F1. Second, excluding the RoBERTa network from CasRel brings noticeable performance loss, with about a drop of 25.6% in F1, a drop of 24.3% in Recall, and a drop of 26.2% in Precision. Based on the above results, we conclude:

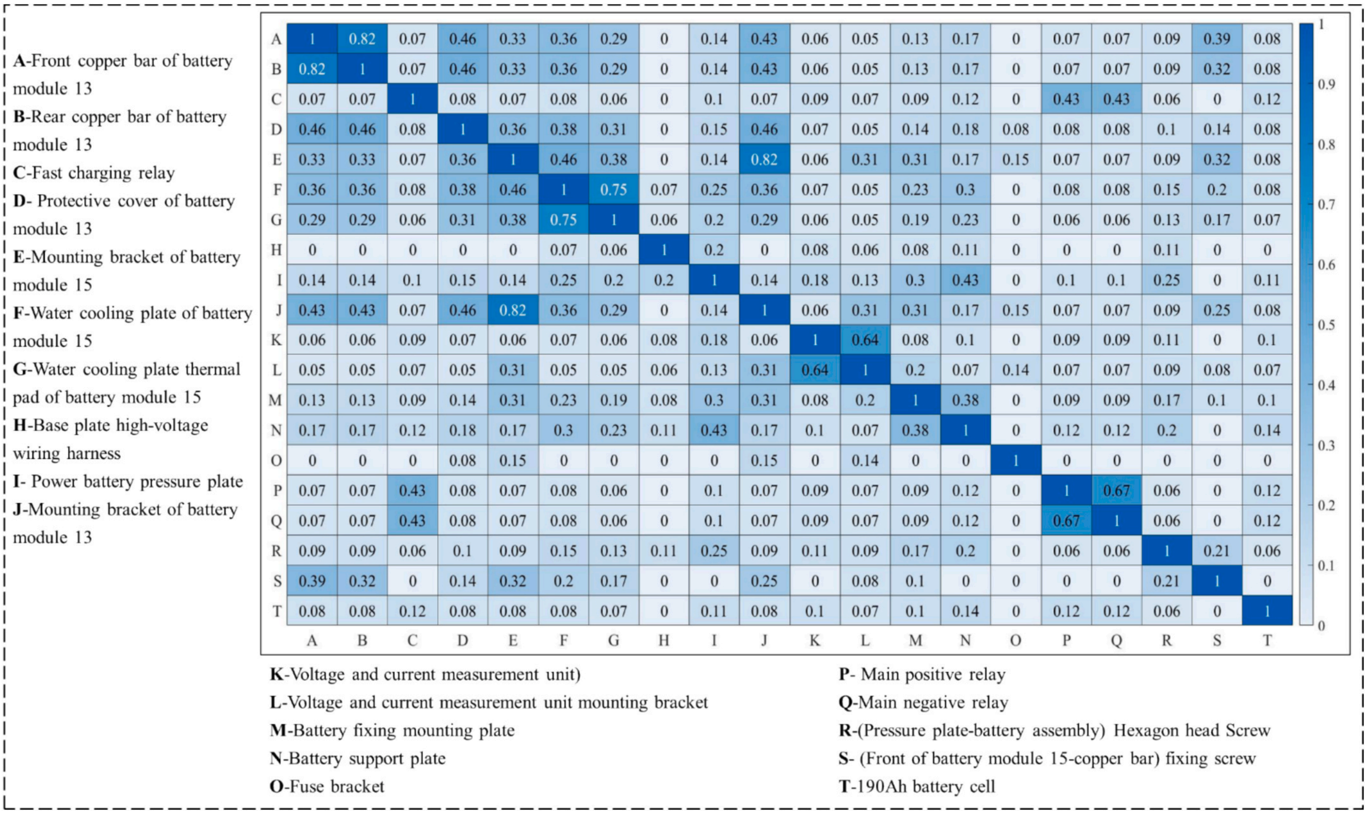


Fig. 8. The calculation results of implicit associations between 20 random nodes.

a) the BiLSTM network improves performance by bidirectionally processing sequential data and capturing contextual dependencies from both past and future elements in the sequence;
b) the RoBERTa network boosts the semantic feature learning ability by generating dynamic word-level representations that allow for more precise comprehension of textual deep meanings.

5.4. Jaccard similarity calculation results

To verify that the proposed method can mine implicit associations between subassembly entities, this study randomly selects 20 entity nodes. By using the Jaccard similarity, we calculate the text similarity to uncover associations between entities. The calculation results are shown in Fig. 8. The depth of the colour in the figure indicates the strength of the association between the corresponding nodes. The darker the colour, the greater the association, and the lighter the colour, the smaller the association. In this study, a similarity threshold of 0.6 is selected, and two nodes with a similarity greater than this threshold can establish an "Association" relation. Node A and node B have a large association value, as they represent distinct components of the same module. Nodes E and J also exhibit a high association. Although they do not belong to the same module, they are components with the same function in different modules. The extraction of these implicit associations is conducive to the development of battery pack automated disassembly. For example, the disassembly processes between node E and node J are similar, and their disassembly experience can be shared, thereby improving the disassembly efficiency.

5.5. ADOKG visualisation and knowledge recommendations

Based on the relational triples obtained in the above chapter, we applied protégé to draw ADOKG, as shown in Fig. 9 (a). By taking the "protective cover of battery module 15" as an example, it can be seen that the protective cover is connected to the "battery module 15" through "4 fixing screws". Therefore, in the actual disassembly process, the first step should involve loosening these four fixing screws. Only after that can the "protective cover of battery module 15" be removed, and then the disassembly of the "battery module 15" can be considered. By calculating the implicit associations between entities, it can be found that there is an association between the "protective cover of battery module 15" entity and the "protective cover of battery module 17" entity. Because they belong to the same category and possess the same function, their disassembly experience and disassembly resources can be shared.

We have conducted an in-depth study on the knowledge recommendations that ADOKG can provide for the automated disassembly process, as shown in Fig. 9.

(1) Operation recommendation. In this study, we recommend that the relation with the "Flexibility" attribute is suitable for manual operation, as flexible robots are expensive and the technology is immature. On the other hand, relations with the "Rigidity" attributes are fit for mechanised operations or automated operations. Robots with rigid structures have made significant progress in both design and application, demonstrating high efficiency and reliability in automated disassembly. Therefore, assigning such tasks to human-robot collaboration or entirely to robots can improve the accuracy and speed of disassembly, and ensure consistency and safety in operations. Specifically, for the physical connections of "Screw fastening" and "Direct connecting", ADOKG recommends the automated operation. In contrast, for the four connections "Wiring harness connecting", "Adhesive bonding", "Snap fastening", and "Cable tie fastening", the manual operation is more suitable. For "Welding" connections, ADOKG recommends the mechanised operation for safety reasons.
(2) The similarity of subassembly entity recommendation. ADOKG can leverage the associations between entities to cluster subassemblies with similar functions, thereby enabling knowledge

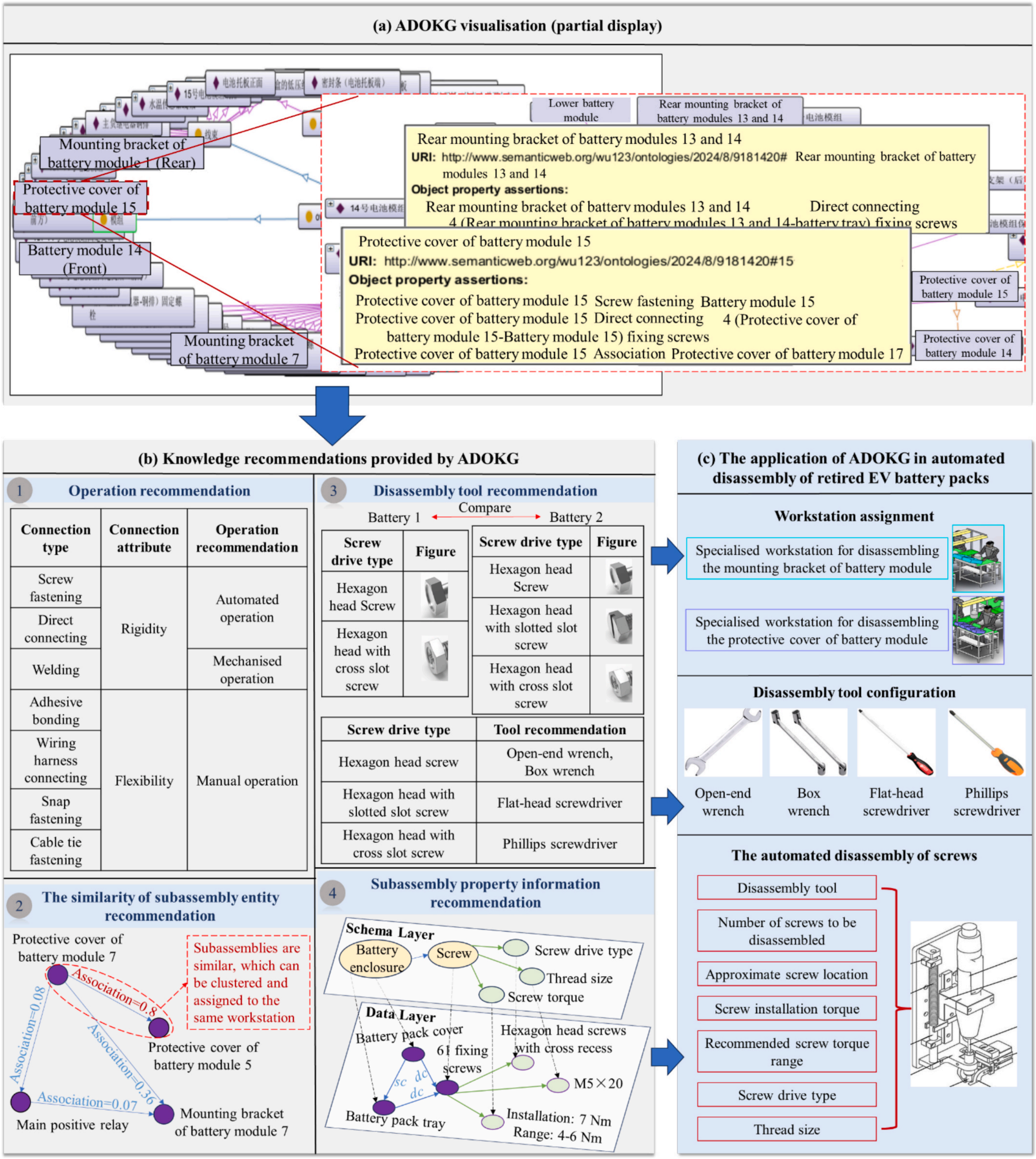


Fig. 9. The knowledge recommendations provided by ADOKG for automated disassembly.

sharing, as well as the assignment and specialisation of disassembly workstations. For instance, the "protective cover of battery module 5" and the "protective cover of battery module 7" can be assigned to the same disassembly workstation, which focuses on disassembling protective covers.

(3) Disassembly tool recommendation. In this study, we enhanced ADOKG by adding screw-matched disassembly tools based on expert experience. For example, for the "61 fixing screws" on the "battery pack cover", the appropriate disassembly tool is Phillips screwdriver. Additionally, it can be identified that there are differences in the drive type of screws between battery 1 and battery 2, with the latter having an additional "hexagon head with cross slot screw". To reduce manual intervention and mitigate machine adjustments or downtime caused by tool mismatches, targeted tools are employed before disassembly, thereby promoting the retired battery pack automated disassembly.

Table 8
The knowledge required for disassembling screws.

Paper No.	Required knowledge	
	Based on the vision	Based on the Force
1	/	Thread size (e.g., M4), Screw drive type (e.g., Torx, external hexagons), Screw Torque (e.g., 3.3Nm)
2	Screw drive type	/
3	Screw drive type, Screw radius	/
4	Screw drive type	/
5	Screw radius	Thread size, Screw drive type, Screw Torque
6	Screw radius	Screw drive type, Screw Torque
7	Screw drive type	Screw Torque
8	Screw drive type	Thread size, Screw Torque, The rotational orientation (e.g., Right)

* "/": The paper does not use any knowledge in this area.

(4) Subassembly property information recommendation. Based on the entity property, ADOKG can provide subassembly information that is crucial for the automated disassembly process. We use screws as an example, which has been widely researched.

Currently, the automated disassembly of screws can be classified based on vision and force perception [3,56–62]. The challenge of vision-based screw removal can generally be attributed to two primary factors: non-frontal orientation and similarity issues, which can lead to misdetection and misclassification. The challenges of force-based automatic screw disassembly lie in the correctness of the tool and the definition of the torque range. As can be seen from Table 8, thread size, screw drive type, screw radius, and screw torque are the basic information for screw automated disassembly. However, information about screws is typically provided or assumed in the previous literature. Under such assumptions (e.g., unknown screws are normally presumed as external hexagon screws), the disassembly process for various screws is thereby simplified to that of a specific type. In actual disassembly scenarios of retired battery packs, information like screw drive type, screw torque, and thread size is often diverse and extracted from relevant files, which undoubtedly impacts the efficiency of the automated disassembly process.

ADOKG can offer rich knowledge against the aforementioned challenges and promote the automation of screw disassembly. First, for a specific subassembly, the number and drive type of screws it is connected to can be provided by ADOKG, thereby minimising misdetection or misdetection. For example, if we know the "battery pack cover" has a "Direct connecting" relation with 61 hexagon head fastening screws, then those 61 hexagon screws must be removed before the "battery pack cover" can be lifted. Secondly, by providing detailed information like thread size and screw torque, ADOKG can support the automated disassembly of screws based on force perception. Additionally, for manual disassembly, ADOKG can reduce unnecessary tool carrying, enhance on-site work efficiency, and assist unskilled workers in performing disassembly tasks.

6. Conclusions

This work studies an automated disassembly-oriented knowledge graph (ADOKG) construction for retired EV battery packs. We employ entities to represent subassemblies and relations to represent explicit physical connections/implicit associations among subassemblies, which can break through the representational limitations of traditional constraint relations and provide reasoning foundations for battery pack automated disassembly systems. Due to massive and unstructured texts of relevant files, compounded by the integrated features of battery pack texts, traditional knowledge extraction methods become ineffective. We propose a candidate entity-based relational triple joint extraction (CE-RTJE) method to finish the disassembly knowledge extraction. In the experiments, more than 10,000 sentences are applied to test CE-RTJE. The experiment results indicate that our proposed approach achieves an F1-score of 93.99% in candidate entity recognition and an F1-score of 95.6% in relational triple extraction. Also, the information of disassembly operations, disassembly tools, and subassembly properties can be recommended by ADOKG for retired battery packs.

However, this study also faces some limitations. For instance, (1) we have not entirely considered the elements of the dismantling site, primarily due to the complexity and variability of disassembly scenarios, as well as the challenges in acquiring original data. Future research should gather more information on the disassembly process to further enhance the knowledge graph. (2) Our data collection on disassembly workers is inadequate, which limits the ability to provide knowledge recommendations for human-robot collaboration in disassembly. Future work should gather detailed information about manual operations through both qualitative and quantitative research methodologies, thereby offering knowledge recommendations for the realisation of effective human-robot collaborative disassembly.

CRediT authorship contribution statement

Yaping Ren: Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Junying Wu:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation. **Cunbo Zhuang:** Writing – original draft, Project administration, Methodology, Funding acquisition, Conceptualization. **Xiaoguang Sun:** Writing – review & editing, Writing – original draft, Resources. **Hongfei Guo:** Writing – original draft, Data curation. **Jianzhao Wu:** Writing – original draft, Methodology, Data curation. **Yang Chen:** Writing – original draft, Data curation. **Jianhua Liu:** Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aei.2025.103525.

Data availability

Data will be made available on request.

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